

Fact Sheet: Leading Causes of Death in VHA Users vs U.S. Population

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CSPEAR provides timely epidemiologic information on VA health care users. This fact sheet presents summary data to inform a broad community of VA leaders, investigators, and clinicians as they consider how best to address the needs of Veterans.

Introduction: The leading causes of death (COD) published annually by the Centers for Disease Control (CDC) National Center for Health Statistics (NCHS) provides a framework for identifying leading COD in the United States. This annual accounting of deaths allows the CDC to monitor trends, recognize emerging challenges, track the effectiveness of interventions, and make public health decisions that improve and save lives ([NVSS - Leading Causes of Death \(cdc.gov\)](#))¹. The prevalence of many diseases such as cancer, diabetes, and mental health have been found to be higher in the U.S. Veteran population than in the overall U.S. population². The objective of this work is to compare and contrast the leading causes of death in the Veterans Health Administration (VHA) user population with the non-VHA user U.S. population.

Figure 1. Mortality Rates, Top 10 Causes of Death, VHA vs CDC, adj: Males (20+)

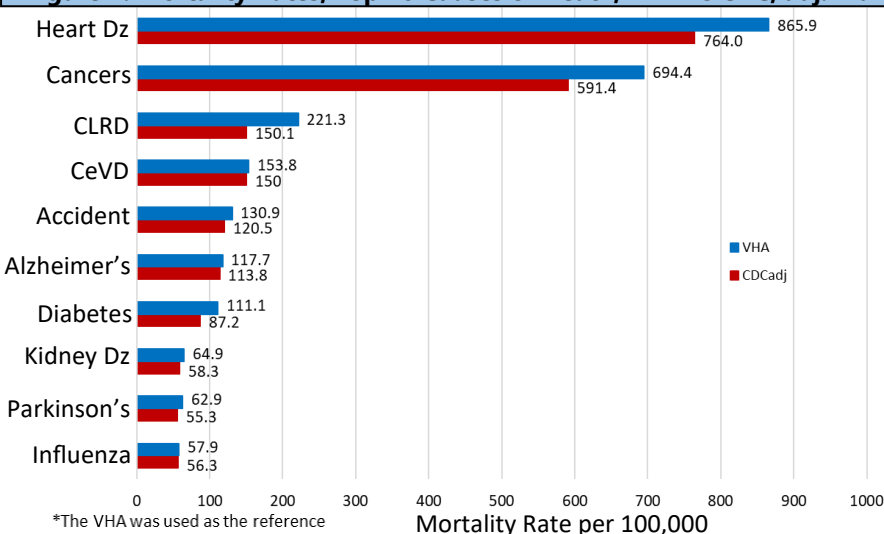
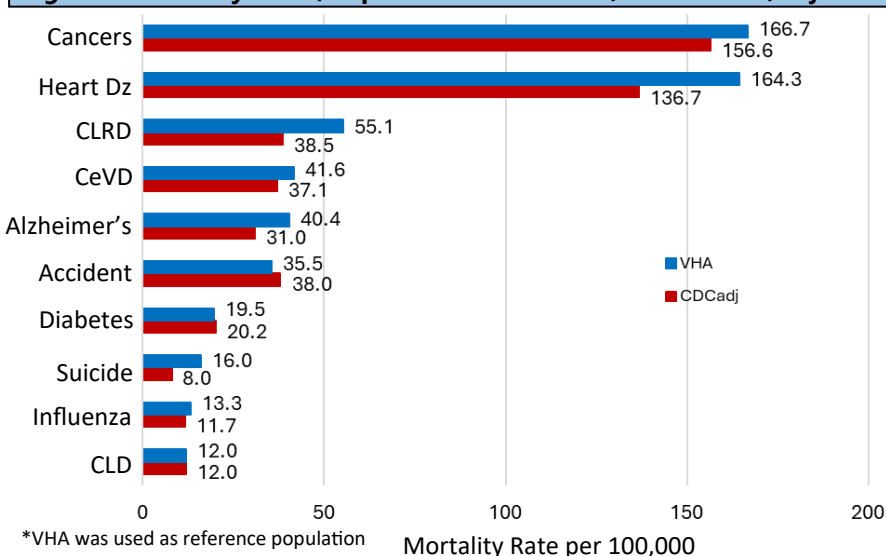


Figure 2. Mortality Rates, Top 10 Causes of Death, VHA vs CDC, adj: Females (20+)



Abbreviations: CDC NCHS: Centers for Disease Control National Center for Health Statistics; CDCadj: Age adjusted (to the VHA reference population) U.S. Population minus VHA user; Heart Dz: Diseases of the heart; Cancers: Malignant Neoplasms; CLRD: Chronic Lower Respiratory Disease; CeVD: Cerebrovascular Disease; Diabetes: Diabetes Mellitus; Kidney Dz: Nephritis, nephrotic syndrome, nephrosis; Influenza: Influenza & Pneumonia; CLD: Chronic Liver Disease and Cirrhosis

Fast Facts

- The 2019 mortality rate for the VHA was 2.7 times the rate in the CDC population (3,033 vs. 1,141 per 100,000) and 1.3 times the CDCadj rate (2,314 per 100,000).
- Among 8,218,677 males and 828,804 females in the VHA and 111,742,810 males and 125,823,816 females in the CDC, the top 10 leading causes of death were similar between the VHA and CDC. However mortality rates were much higher for the VHA population. For VHA Males: VHA/CDCadj Rate Ratio (RR) for Heart Dz was 1.13, Cancer (1.17), CLRD (1.47), Diabetes (1.27). For VHA females Heart Dz was 1.20, Cancer (1.06), CLRD (1.43), Alzheimer's (1.30) and Suicide (2.00).
- VHA male and female suicide mortality rates were higher than CDC adjusted rates in every age category except 85+ (Figure 3). VHA males peaked in 25-34 and 85+, while the CDC had lower rates in 25-74 and peaked in the 85+ age group. Trends for females were similar between the VHA and CDC adjusted rates and peaked in the 45-54 age group.
- VHA had higher rates for Heart Dz across both sexes for ages 55+ groups, the largest was for females 85+ (RR=1.2).
- VHA males had higher rates for cancer in the 65-84 age groups (RR=1.2-1.4) and VHA females had higher rates in those aged 55+ (RR=1.1-1.3).

Visit [CSPEAR's website](#) or contact CSPEAR@va.gov for more information.

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Veterans Health Administration
Cooperative Studies Program

Figure 3. VHA vs CDC Mortality Rates per 100,000 in 2019 by Sex for Selected Conditions

Figure 3a. Suicide Death Rates for Males

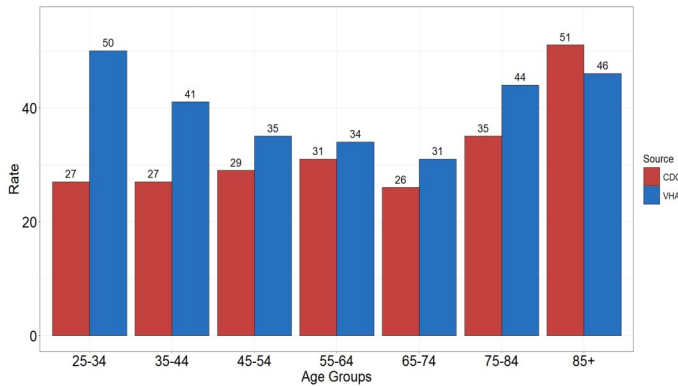


Figure 3b. Suicide Death Rates for Females

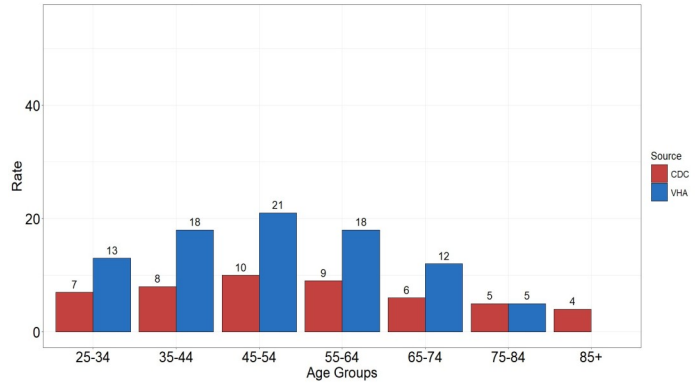


Figure 3c. Heart Disease Death Rates for Males

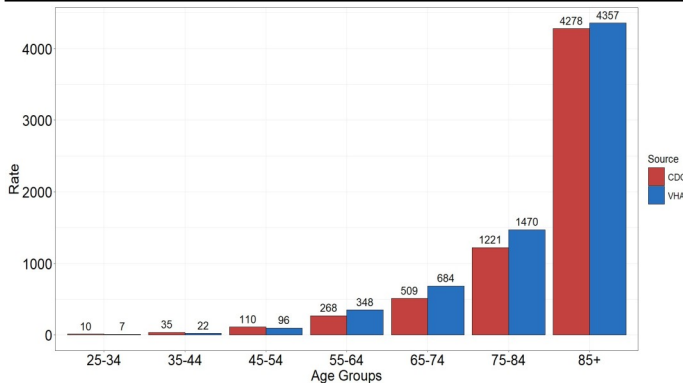


Figure 3d. Heart Disease Death Rates for Females

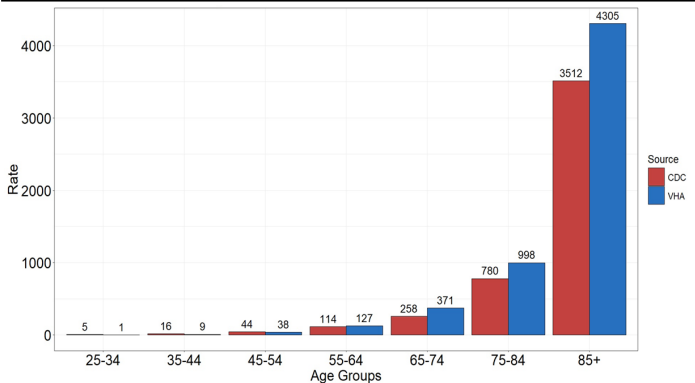


Figure 3e. Cancer Death Rates for Males

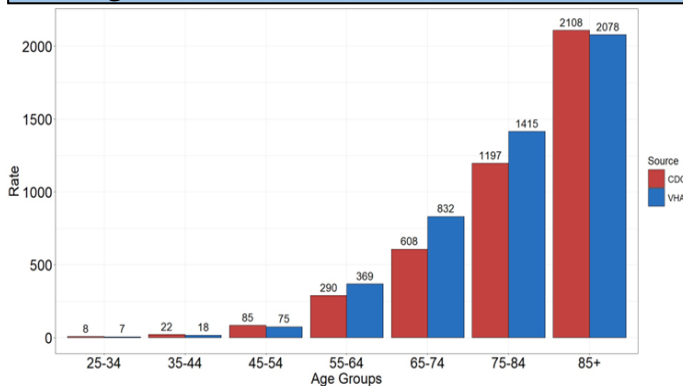
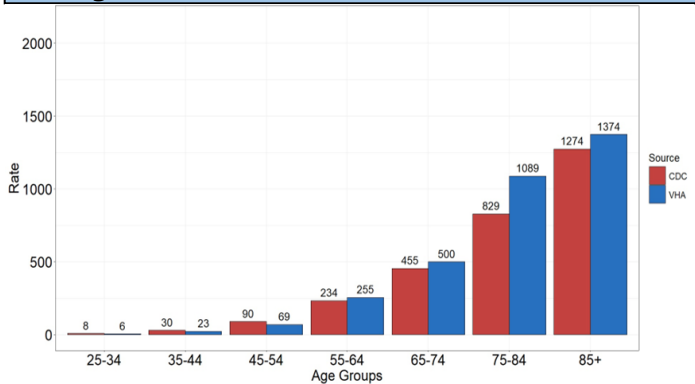


Figure 3f. Cancer Death Rates for Females



Methods:

Data were extracted from the Corporate Data Warehouse (CDW), a national database that integrates clinical and administrative information in the VHA. All VHA users were identified who had received any inpatient or outpatient care at a VHA facility any time during 2004 through 2018, were between 20 and 110 years old, had been assigned into 1 of the 8 VHA priority groups and were alive in 2019.

Dates of death and the underlying cause of death for Veterans who had died during that same period were received from the National Death Index (NDI) which provides mortality data from death certificates. CDC data were obtained from the [CDC WONDER](https://wonder.cdc.gov/) leading causes of death database. Causes of death in 2019 were grouped and ranked according to CDC, NCHS developed phenotypes¹. Results are presented overall and by age and sex.

VHA users are a subset of the United States population with many more older adults (i.e. ≥ 65 : 48% vs 21%). Therefore, we compare VHA user mortality rates with the CDC population minus the VHA user population (CDC-VHA) and use direct standardization techniques to age-adjust the CDC population estimates (CDCadj). The VHA user age distribution was used as the reference population for standardization.

Notes: This work was conducted under CSPEAR's operational access to VA data. This material is the result of work supported with resources and the use of facilities at the VA Cooperative Studies Program Epidemiology Center in Boston, MA. The contents do not represent the views of VA or the US Government. Additional VA statistics are provided in the Supplement.

References and Resources

1. Heron M. Deaths: Leading causes for 2019. National Vital Statistics Reports; vol 70 no 9. Hyattsville, MD: National Center for Health Statistics. 2021. DOI: <https://dx.doi.org/10.15620/cdc:107021>
2. Eibner C et al. Current and Projected Characteristics and Unique Health Care Needs of the Patient Population Served by the Department of Veterans Affairs. Rand Health Q. 2016 May 9;5(4):13. PMID: 28083423; PMCID: PMC5158228.